Advanced Computer Networks

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Analysis of 1-persistent CSMA

DSSRG: Decentralized Smart Systems Research Group

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Performance Analysis of the *unslotted 1-persistent* CSMA protocol

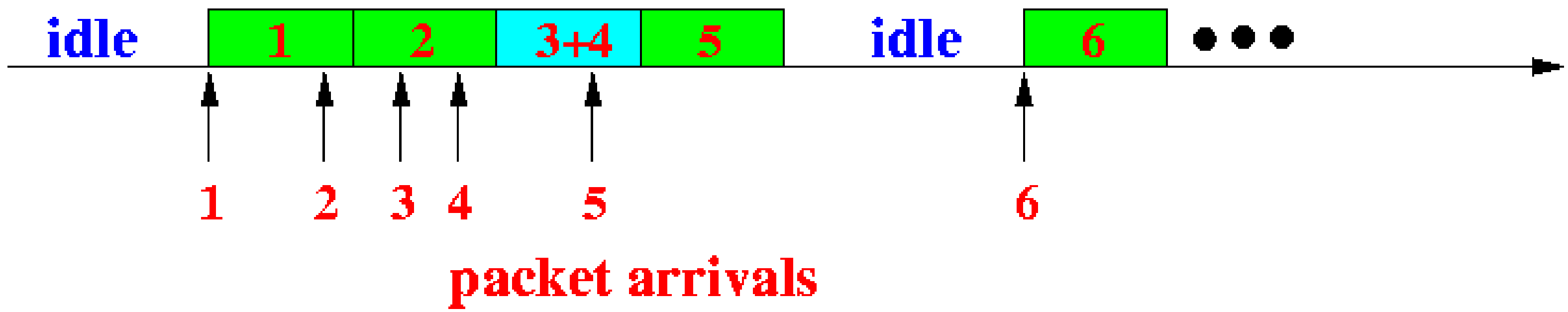
- **Some observations about the 1-persistent CSMA protocol**

- **A quick refresher of the 1-persistent CSMA protocol:**
- When a **node** senses that the **channel is busy**, it will **wait for the end of the transmission** and **transmit** ***immediately***

- **Consequence of this fact:**
- **All nodes that sense the channel is busy (during some node's transmission) will transmit their packets at the end of the current transmission**

Transmission pattern in the 1-persistent CSMA protocol:

- When observing the **1-persistent CSMA** protocol, you will see the following **transmission pattern**:



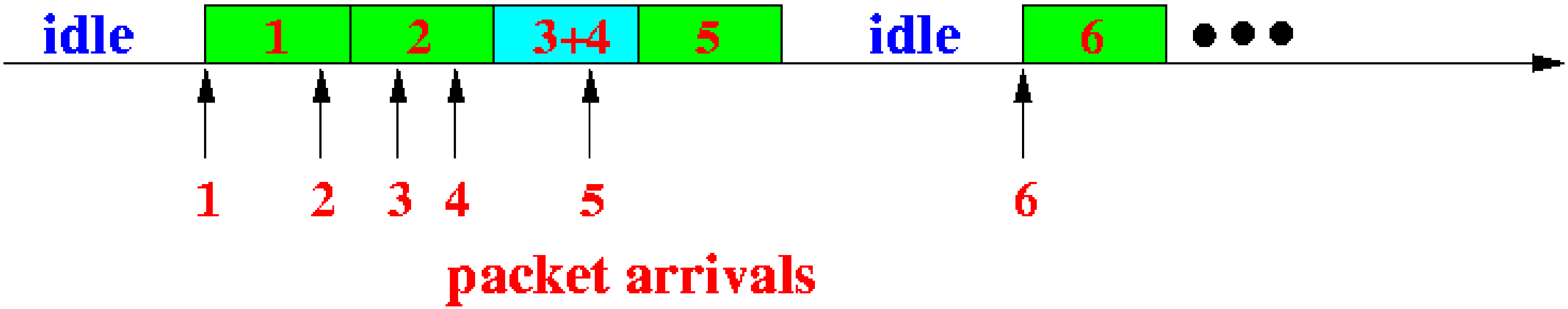
- There are again a series of **interleaved busy periods** and **idle periods**
- Each cycle starts with an **idle period**, **followed** by a **busy period**

Pattern

- There are again a series of **interleaved busy periods** and **idle periods**
- Each cycle starts with an **idle period**, **followed** by a **busy period**
- A **busy period** in the **1-persistent CSMA protocol** consists of **one or *more* transmission periods**
- (A **busy period** in the **non-persistent CSMA protocol** consists of **exactly *one* transmission period.** This is because when a **node** senses that the **channel is busy** in the **non-persistent CSMA protocol**, it will **leave the queueing system**)

Anatomy of the *1-persistent* CSMA busy period:

- The **transmission periods** inside one **busy period** can be of the following type:



Major difference:

In 0-persistent when one occupies the rest leaves the queue,

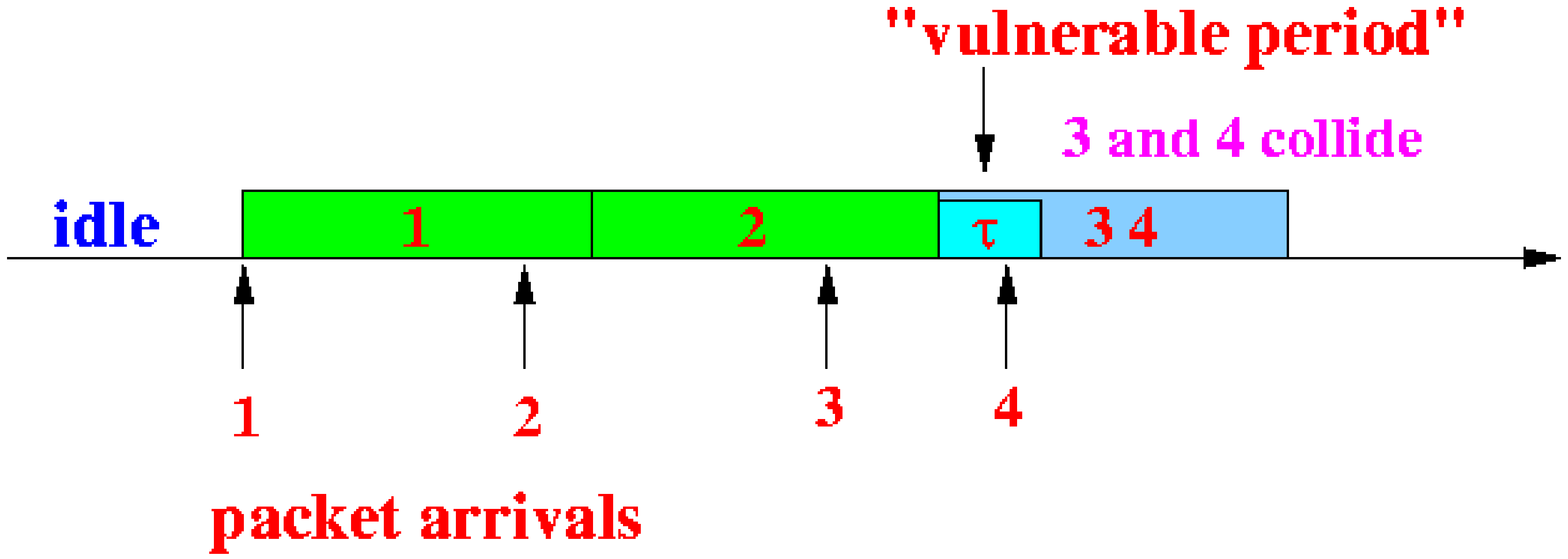
while in 1-persistent – they don't leave – Rather stick to it

There are two different types of transmission periods inside a busy period in 1-p

- The **first type** is a **transmission period** where the **number of packet arrivals at the *start*** of the **transmission period** is equal to **1** → We call this a **type 1 period**

- **Type 1 periods can *become* useful**

- A **type 1 period can** contain a **collision** when **another packet** arrives **during the *vulnerable period***:


"vulnerable period"
3 and 4 collide
idle
1
2
τ
3 4
1
2
3
4
packet arrivals

Note

•**All three periods** in the **figure** are **type 1 periods**

•There is **only one packet** arrival at the **start** of the period

•The **period #1 and #2** are **useful periods** because the packet arrival **will *not* cause a *collision***

•The **period #3** is ***not* useful** because the **arrival of packet 4 will cause a *collision***

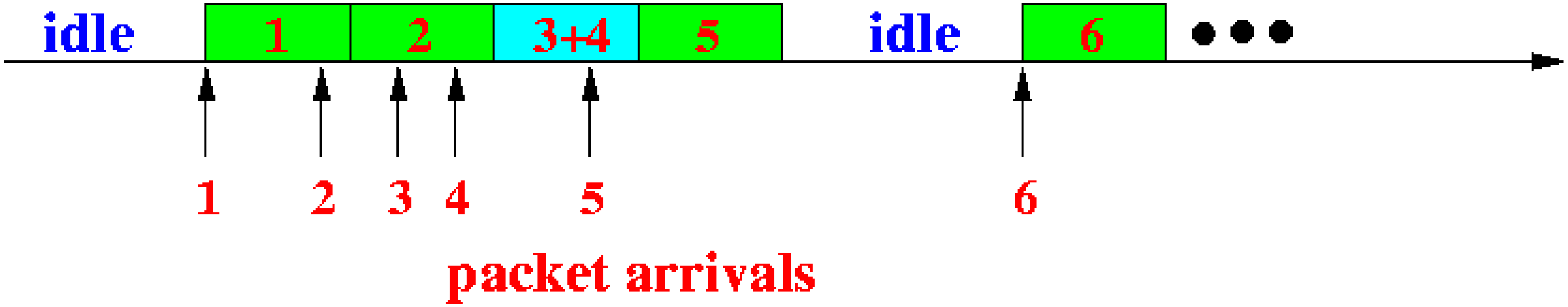
•The **first transmission period** in a **busy period** is **always** a **type 1 period** (Because there are **no *simultaneous* arrivals** in a **Poisson Arrival process**)

Note

- The Poisson distribution and Poisson process are designed to model independent, continuous-time events where the probability of two or more events occurring at the exact same instant is zero.

- **This is a foundational, mathematical assumption rather than a universal law of nature.**

- The **other type** is a **transmission period** contains a transmission of **2 or more packets**
- We call this a **type 2 period**, **Example**



- A **type 2 period** is started when there are **2 or more arrivals** during the ***previous*** **transmission period**
- **Type 2 periods** are **certainly *not* useful**
- (because there ***will*** **be a collision** in the transmission period)

- According to the **same principle** (counting the **number of arrivals** in the ***previous*** **transmission period**), we see that:

- An **idle period** is started when there is **0 arrival** during the ***previous*** **transmission period**

- We will use a similar nomenclature for the **idle period**:

- An **idle period** is called a **type 0 period**

Type of of periods –

- Type 0: **No one attempts: Idle period**

- Type 1: **Only one attempts in the previous period**
- **No collision** – if only one attempts
- **Collision - Attempt in vulnerable period**

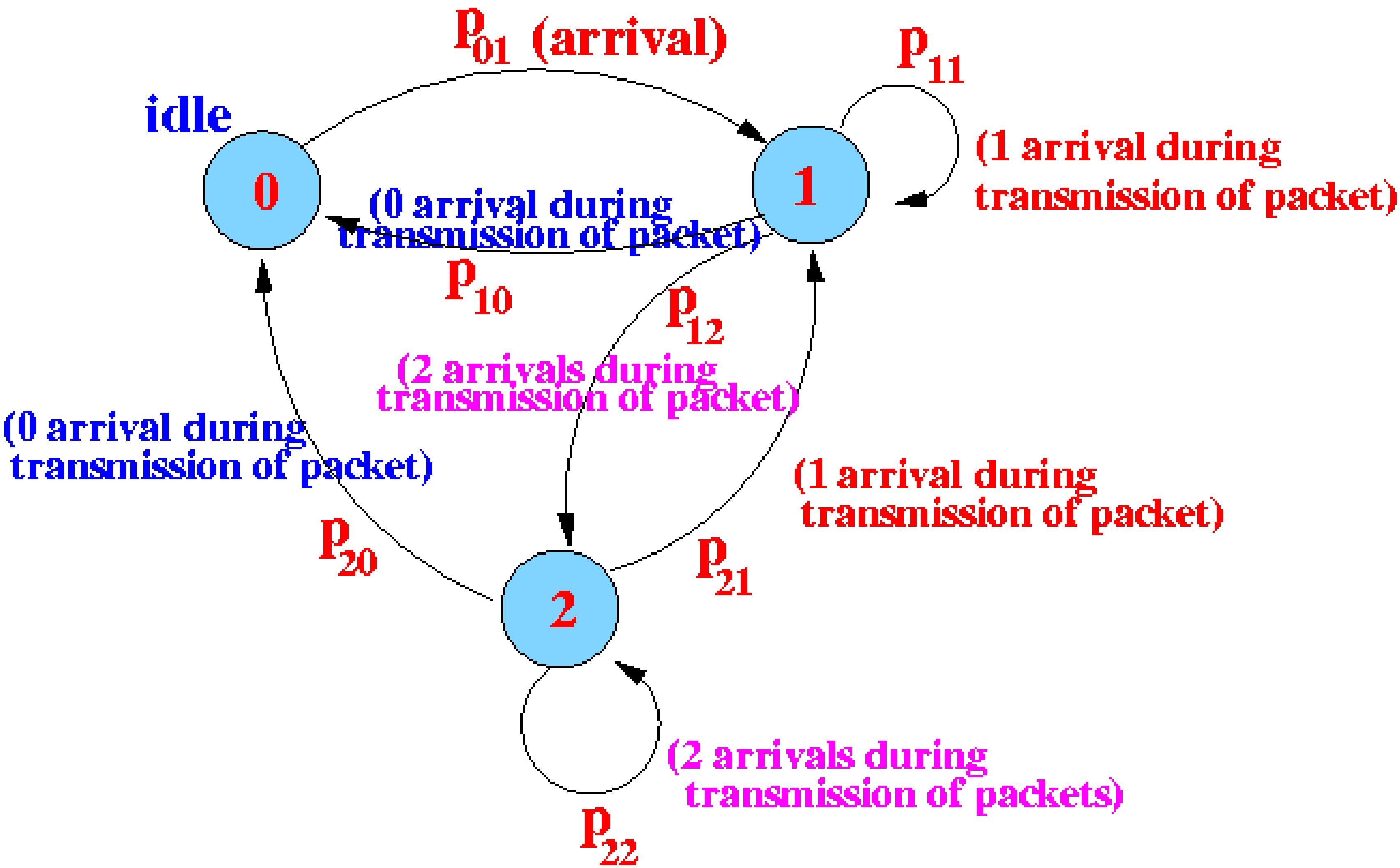
- Type 2: **More than one attempts in the previous period: Collision -**

Markov transition diagram for the 1-persistent CSMA protocol

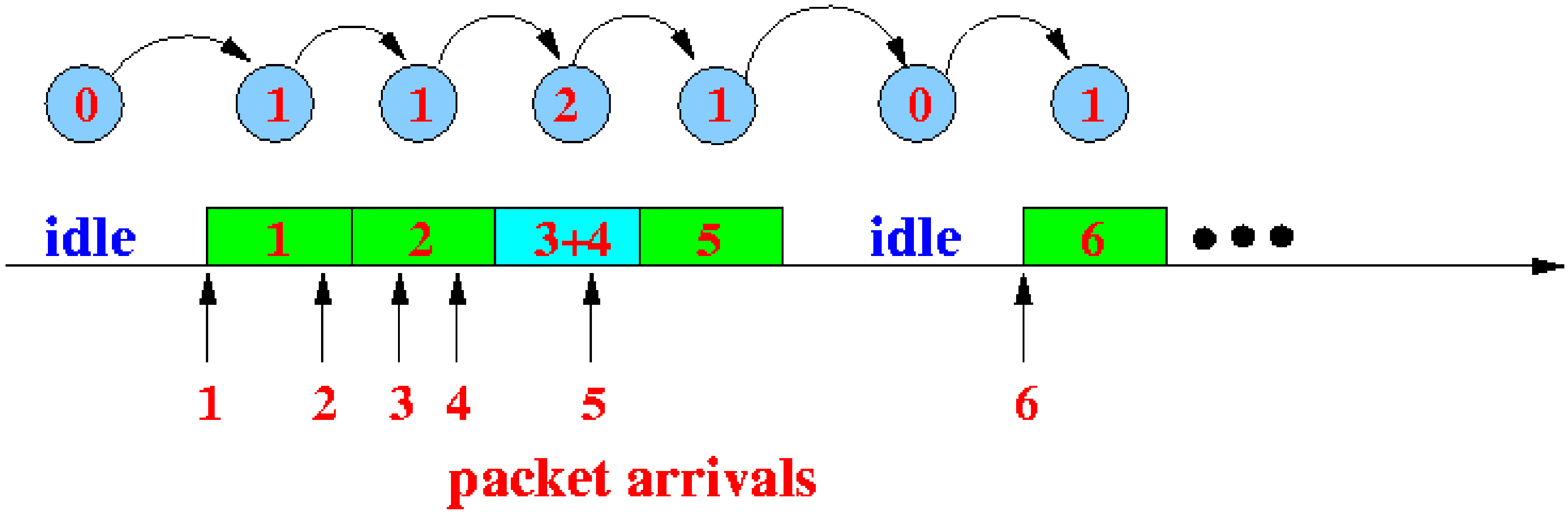
- **Definitions of the different states:**

- **State 0** = the **number of packet arrivals at the** ***start*** of the **transmission period** is equal to **0**
- **State 1** = the **number of arrivals at the** ***start*** of the **transmission period** is equal to **1**
- **State 2** = the **number of arrivals at the** ***start*** of the **transmission period** is equal to **2 (or more)**

The **state transition diagram** of the **1-persistent (unslotted) CSMA** protocol:



Example state transitions:



Throughput of the unslotted 1-persistent CSMA protocol

Definitions and notations:

- π_i = **steady state probability** of the **system** in **state i**
- **(for i = 0, 1, 2)**

- T_i = **random variable** representing the **length (duration)** of a **type i period (for i = 0, 1, 2)**
- $\underline{T}_i$ = **E[T]**
- (i.e., the **average length (duration)** of a **type i period (for i = 0, 1, 2)**

Transmission Time and Duration of the stages

- Do not **confuse** the **random variable $\underline{T}_i$** with the **constant T**

- **Recall** that:

- **T = length (duration)** of a **packet transmission**
- This value is **constant**

Throughput of the unslotted 1-persistent CSMA protocol:

Throughput will be the ratio of the time period which could be

[Successfully used for packet transmission]

And

[The total time period]

Calculate - Total time period

Average duration (length) of *one* transmission period

$$\pi_0\underline{T}_0 + \pi_1\underline{T}_1 + \pi_2\underline{T}_2$$

40% of the time, the system is in **state 0,** It lasts approximately **5 min**
50% of the time, the system is in **state 1,** It lasts approximately **1 min**, and
10% of the time, the system is in **state 2,** It lasts approximately **2 min**,

Average **duration** of **one period** is:

$$0.4 \times 5 + 0.5 \times 1 + 0.1 \times 2$$

- From the discussion above we know that – The system can stay in three types of periods State 0, State 2 and State 3.

- Steady-state probabilities shows the **probability** that the system will stay in a given state
- Length of a state will tell us how much time is the **duration** of a state in terms of time.

Average total time is = **Sum over all states [probability x duration]**

- To calculate the throughput we need some base line to compare the successful part –
- This is the part we compare with the successful part …

Calculate - useful time period

- U*seful* (successful packet transmission)
- $\pi_i T \times$ Ᵽ[success in T_i]

$\pi_0 T \times$ Ᵽ[success in T_0] +

$\pi_1 T \times$ Ᵽ[success in T_1] +

$\pi_2 T \times$ Ᵽ[success in T_2]

Summary

$$S = \frac{\pi_0 T \times \mathbb{P}[\text{ success in } T_0 \text{ }] + \pi_1 T \times \mathbb{P}[\text{ success in } T_1 \text{ }] + \pi_2 T \times \mathbb{P}[\text{ success in } T_2 \text{ }]}{\pi_0 \underline{T}_0 + \pi_1 \underline{T}_1 + \pi_2 \underline{T}_2}$$

$$\frac{\pi_0 T \times 0 + \pi_1 T \times \mathbb{P}[\text{ 0 arrival in } \tau \text{ sec }] + \pi_2 T \times 0}{\pi_0 \underline{T}_0 + \pi_1 \underline{T}_1 + \pi_2 \underline{T}_2}$$

Because success **is possible only in State 1** – That is also if and only if there is no arrival in the ***vulnerable period***

What remains to be done:

- Compute the **average durations** $\underline{T}_0$, $\underline{T}_1$ and $\underline{T}_2$

- And compute the **steady state probabilities** π_0, π_1 and π_2

One by one ….

$\underline{T}_0$: The average length of a *type 0* period

- The **type 0 period** is an **idle period**
- $\underline{T}_0$ = **average duration** of an **idle period**

- The **duration** of an **idle period** depends **only** on the **arrival process**

- Since the **arrival process** is a **Poisson process** with **arrival rate g**, the **duration** of an **idle period** in **1-persistent CSMA** is the **same** as the **duration** of an **idle period** in ***non*-persistent CSMA**

Therefore:

$$T_0 = \frac{1}{g}$$

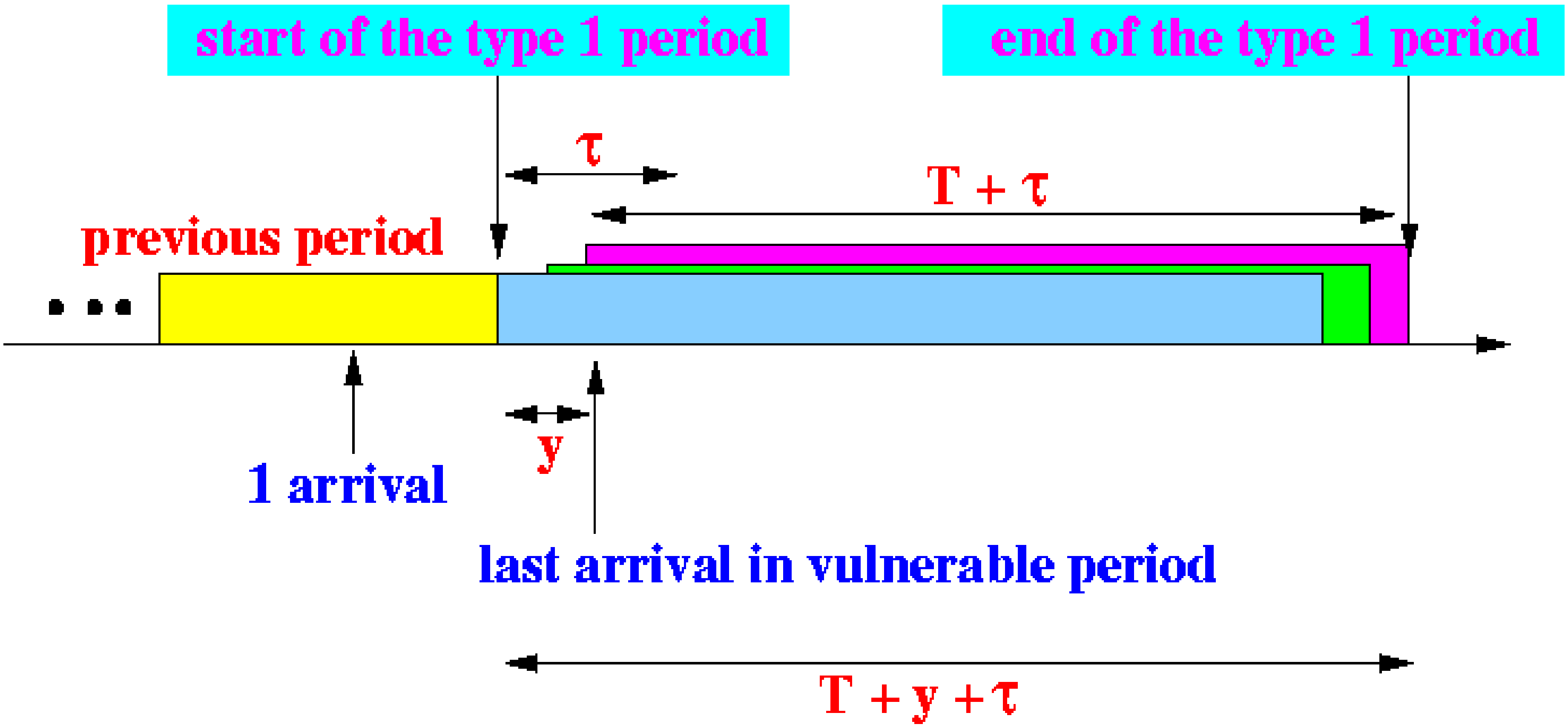
(The same as we derived earlier … Recall)

$\underline{T}_1$: The average length of a *type 1* period

- **Although** there is **one arrival** at the **start** of a **type 1 period**, it is **not true** that a **type 1 period** will **always** contain **one packet transmission**

- The **number of packets transmitted** in a **type 1 period** depends on the **number of arrivals during the** ***vulnerable period*** of the **initiating transmission**

Anatomy of a type 1 transmission period:



- **y** = a **random variable** representing the **arrival time** of the **last arriving transmission** in the **vulnerable period**

- **We have that:** $\mathbf{T_1 = T + \tau + y}$
- Thus, $E[T_1] = T + \tau + E[y]$

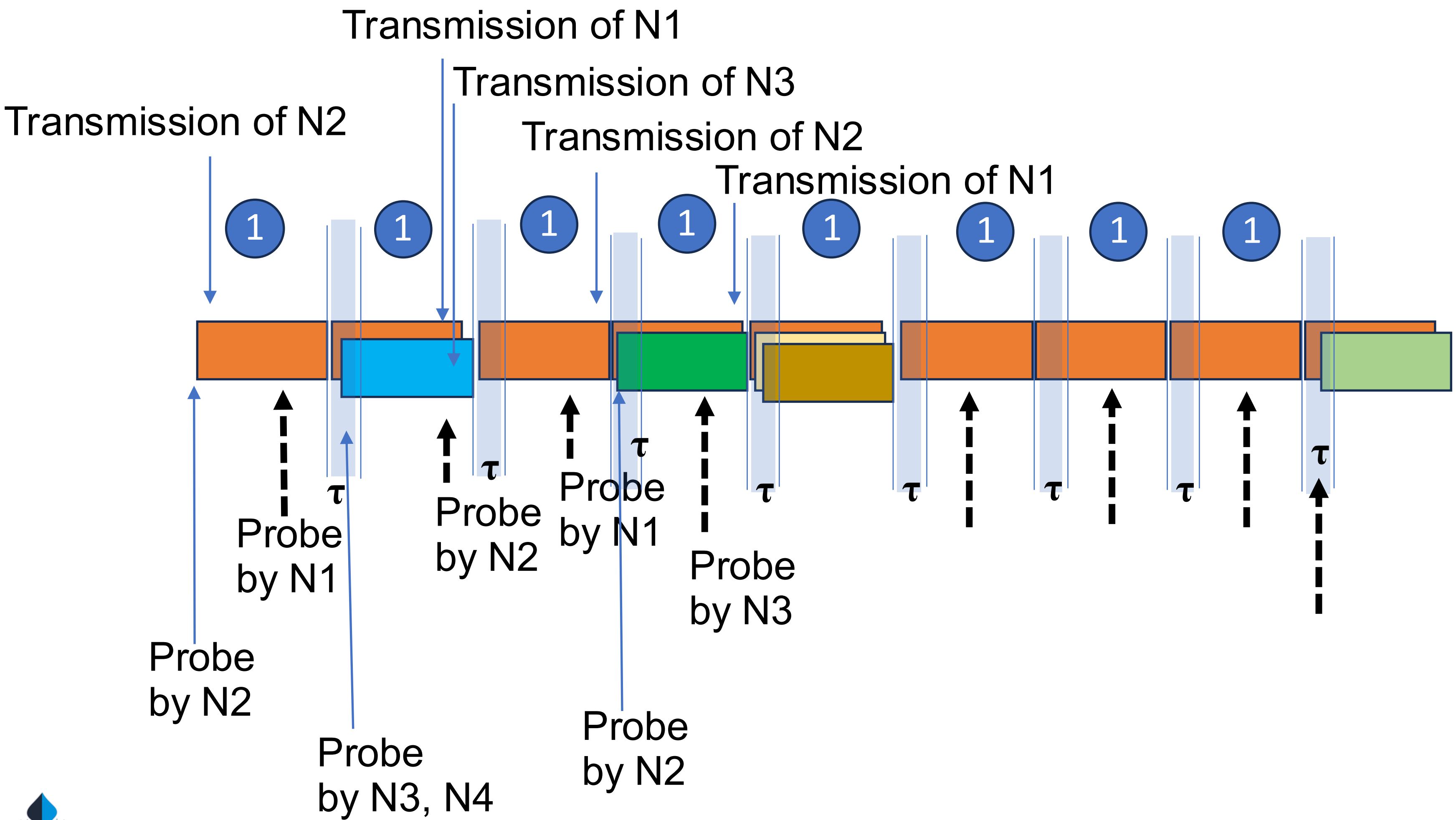
- The **random variable y** (arrival time of the last arriving packet) depends **only** on the **arrival process**
- For a **Poisson arrival process** with **arrival rate g**, we have **previously determined** that

$$y = \tau - \frac{1 - e^{-g\tau}}{g}$$

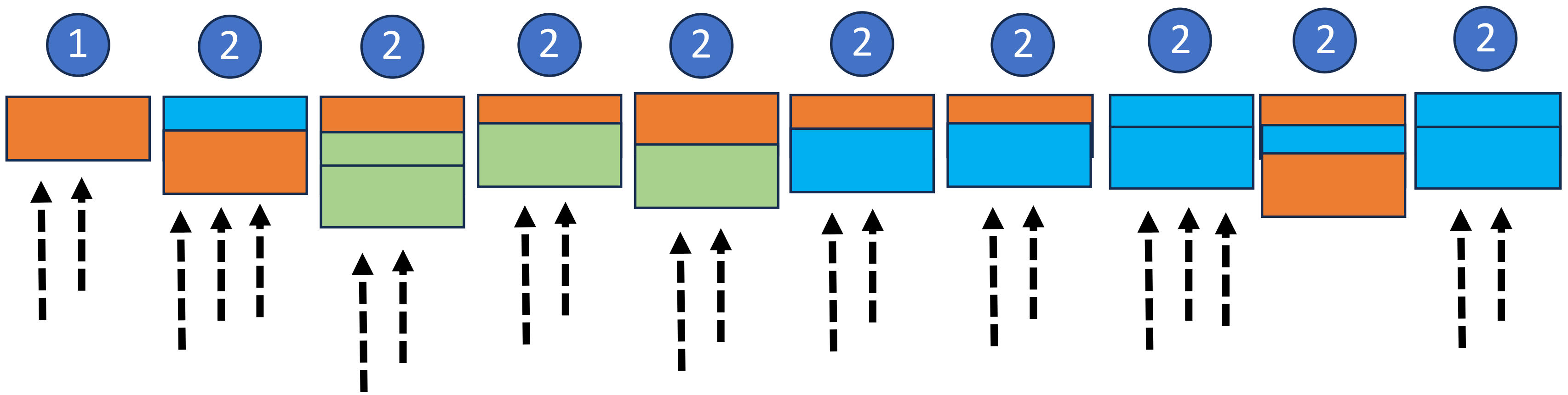
$$E[T_1] = T + \tau + E[y]$$

$$T_1 = T + 2\tau - \frac{1 - e^{-g\tau}}{g}$$

T1: Imagine there are four nodes N1, N2, N3, N4

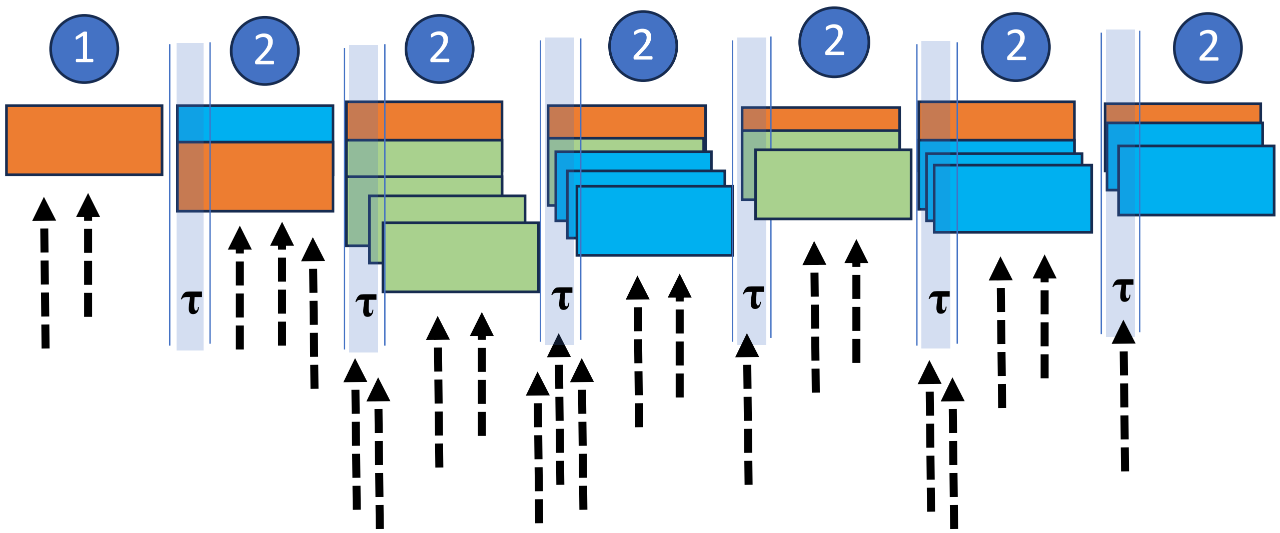


T2: Consider the following scenario --



T2: Consider the following scenario –

What is the difference?



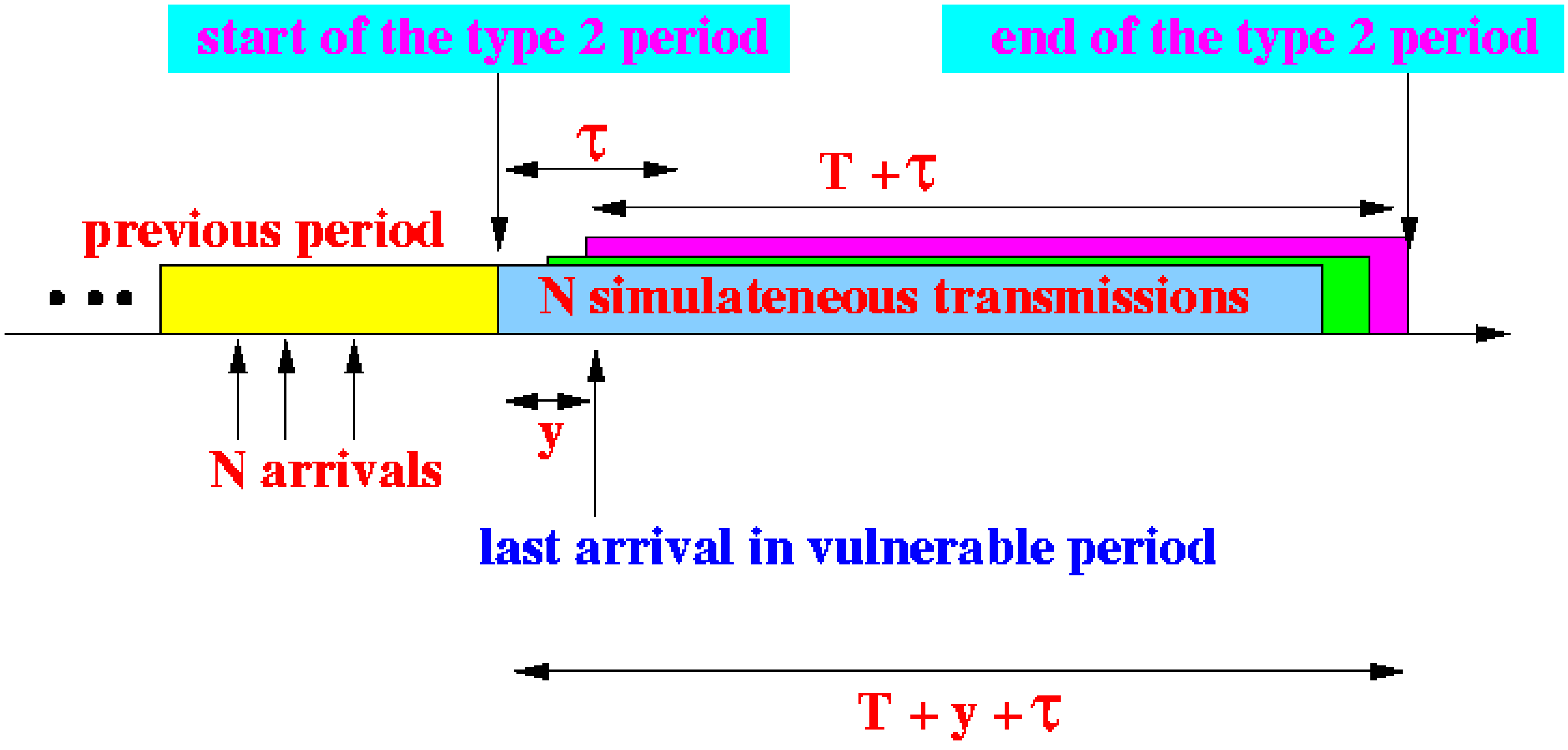
$\underline{T}_2$: The average length of a *type 2* period

- **Similar** to the **type 1 period**:

- **Although** there are **N arrivals** at the **start** of a **type 2 period**, it is ***not*** **true** that a **type 2 period** will **always** contain **N packet transmissions**

- The **number of packets transmitted** in a **type 2 period also** depends on the **number of arrivals during the *vulnerable period*** of the **initiating transmissions**

Anatomy of a type 2 transmission period:



- **y** = a **random variable** representing the **arrival time** of the **last arriving transmission** in the **vulnerable period**

- **We see that:** The **duration** of a **type 2 period** is **identical** to the **duration** of a **type 1 period** !!!

- $E[T_2] = E[T_1]$

$$T_2 = T + 2\tau - \frac{1 - e^{-g\tau}}{g}$$

Status …

- We got expressions for $\mathbf{T_0}$, $\mathbf{T_1}$ and $\mathbf{T_2}$....
- Now we find the **steady state probabilities** $\boldsymbol{\pi_0}$, $\boldsymbol{\pi_1}$ and $\boldsymbol{\pi_2}$....
- **In other words what are the chances that the system can be in state 0 state 1 and state 2.**

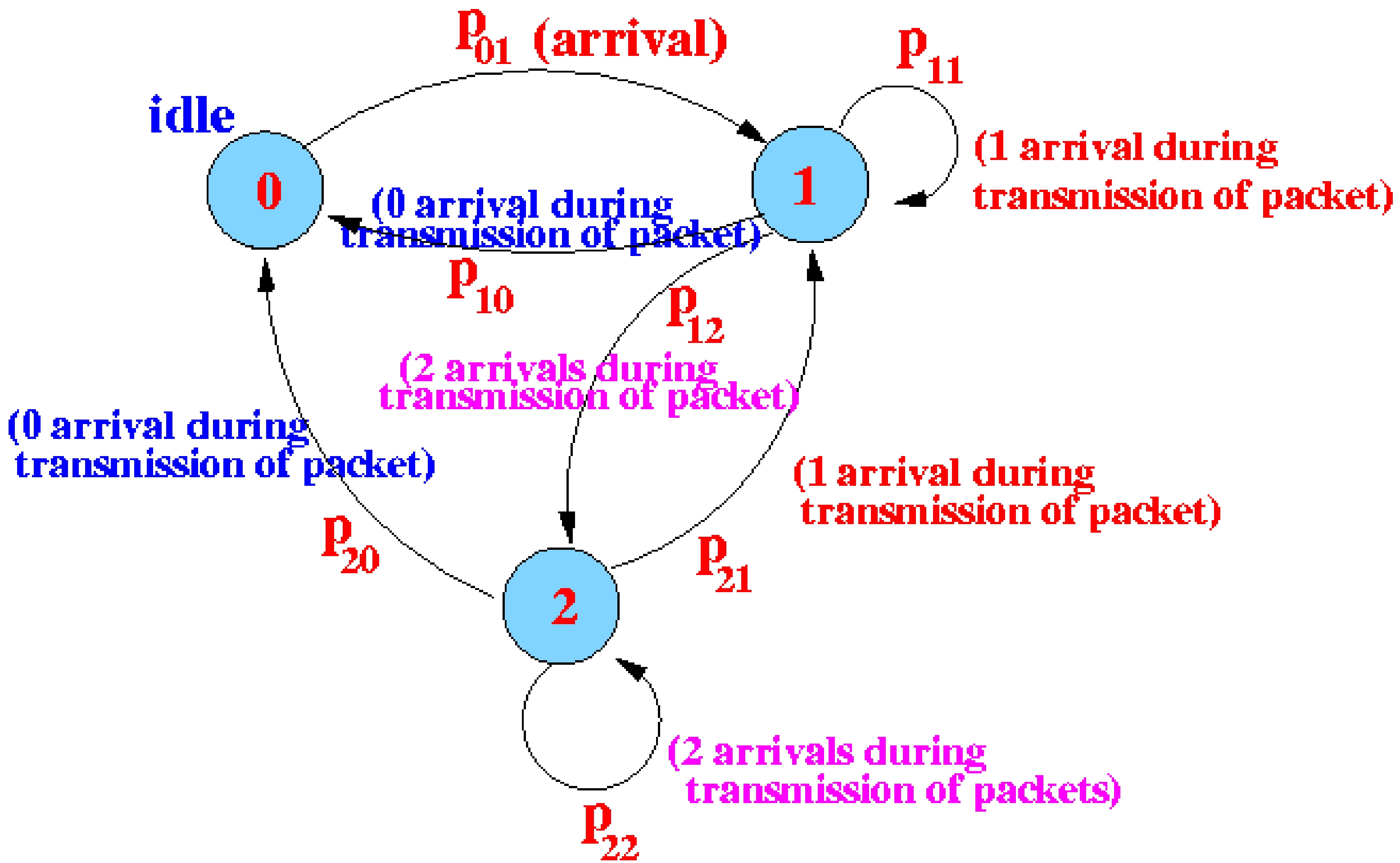
Steady state probability π_0, π_1 and π_2

- Here we **apply** the material from **Markov chain**....

- However, before we can apply the **Markov analysis**, we must **first determine** the **transition probabilities p_{00}, p_{01}, p_{02}, p_{10},** …

- Unlike **artificially** created **Markov chains** with **chosen transition probabilities** [e.g., in our examples!...] :
- **Real-life systems** have **transition probabilities** that must be **discovered** by **analysis**

The state transition diagram:



Steady state probabilities -

- The **steady state probabilities π_0, π_1, π_2** are computed **using the following equations [in general]**

- $\pi_0 = \pi_0 p_{00} + \pi_1 p_{10} + \pi_2 p_{20}$

- $\pi_1 = \pi_0 p_{01} + \pi_1 p_{11} + \pi_2 p_{21}$

- $\pi_2 = \pi_0 p_{02} + \pi_1 p_{12} + \pi_2 p_{22}$

From the diagram:

- We have: $p_{00} = 0$ and $p_{02} = 0$
- Substituting: $p_{00} = 0$ and $p_{02} = 0$ in the equations:
- $\pi_0 = \pi_1 p_{10} + \pi_2 p_{20}$
- $\pi_1 = \pi_0 p_{01} + \pi_1 p_{11} + \pi_2 p_{21}$
- $\pi_2 = \pi_1 p_{12} + \pi_2 p_{22}$

Steady state probabilities

- From: $p_{00} + p_{01} + p_{02} = 1$
- and: $p_{00} = 0$ and: $p_{02} = 0$

- We have: $p_{01} = 1$, So, substitute $p_{01} = 1$ we get:

- $\pi_0 = \pi_1 p_{10} + \pi_2 p_{20}$
- $\pi_1 = \pi_0 + \pi_1 p_{11} + \pi_2 p_{21}$
- $\pi_2 = \pi_1 p_{12} + \pi_2 p_{22}$

Further simplification:

- The **transition probability** from **state 1 to state 0** **(p_{10})** depends on the **number of arrivals** in a **type 1 period**
- **Specifically: state 0** is ***reached*** when there are **no arrivals** within the **type 1 period**

- The **transition probability** from **state 2 to state 0** **(p_{20}) also** depends on the **number of arrivals** in a **type 2 period**
- **Specifically: state 0** is ***reached*** when there are **no arrivals** within the **type 2 period**

Recall that:

- The **duration** of a **type 1 period** and a **type 2 period** have the ***same*** **probability distribution** !!!

- Because the **arrival process** to **type 1 and type 2 periods** are **also identical (Poisson with rate g)**, we conclude that:

- $\mathbf{p_{10} = p_{20}}$ (Why?? Imagine the scenarios…)

- The **transition probability** from **state 1 to state 1 (p_{11})** depends on the **number of arrivals** in a **type 1 period**
- **(Specifically: state 1** is ***reached*** when there are **exactly 1 arrival** within the **type 1 period**)

- The **transition probability** from **state 2 to state 1 (p_{21}) also** depends on the **number of arrivals** in a **type 2 period**

- **(Specifically: state 1** is ***reached*** when there are **exactly 1 arrival** within the **type 2 period**)

Recall again that:

- The **duration** of a **type 1 period** and a **type 2 period** have the ***same*** **probability distribution** !!!

- Because the **arrival process** to **type 1 and type 2 periods** are **also identical (Poisson with rate g)**, we conclude that:

- $\mathbf{p_{11} = p_{21}}$

- The **transition probability** from **state 1 to state 2 (p_{10})** depends on the **number of arrivals** in a **type 1 period**

- **(Specifically: state 2** is ***reached*** when there are **more than 2 arrivals** within the **type 1 period**)

- The **transition probability** from **state 2 to state 0 (p_{20}) also** depends on the **number of arrivals** in a **type 2 period**
- **(Specifically: state 2** is ***reached*** when there are **more than 2 arrivals** within the **type 2 period**)

- **Same old story again, recall that:**

- The **duration** of a **type 1 period** and a **type 2 period** have the ***same*** **probability distribution** !!!

- Because the **arrival process** to **type 1 and type 2 periods** are **also identical (Poisson with rate g)**, we conclude that:

- $\mathbf{p_{12} = p_{22}}$

Result after substitution:

- Before substitution:
- $\pi_0 = \pi_1 p_{10} + \pi_2 p_{20}$
- $\pi_1 = \pi_0 + \pi_1 p_{11} + \pi_2 p_{21}$
- $\pi_2 = \pi_1 p_{12} + \pi_2 p_{22}$

- After substitution:
- $\pi_0 = \pi_1 p_{10} + \pi_2 p_{10}$
- $\pi_1 = \pi_0 + \pi_1 p_{11} + \pi_2 p_{11}$
- $\pi_2 = \pi_1 p_{12} + \pi_2 p_{12}$

- We must **replace *any* one of the equations** with:
- $\boldsymbol{\pi_0 + \pi_1 + \pi_2 = 1}$

- $\pi_0 = \pi_1 p_{10} + \pi_2 p_{10}$
- $\pi_1 = \pi_0 + \pi_1 p_{11} + \pi_2 p_{11}$ ====> $\pi_0 + \pi_1 + \pi_2 = 1$ **(Replace)**
- $\pi_2 = \pi_1 p_{12} + \pi_2 p_{12}$

We have the set of equations
(After simplifications)

$$\pi_0 = \pi_1 p_{10} + \pi_2 p_{10}$$

$$\pi_2 = \pi_1 p_{12} + \pi_2 p_{12}$$

$$\pi_0 + \pi_1 + \pi_2 = 1$$

Solving π_0, π_1, π_2:

$$\pi_0 = \frac{p_{10}}{1 + p_{10}}$$

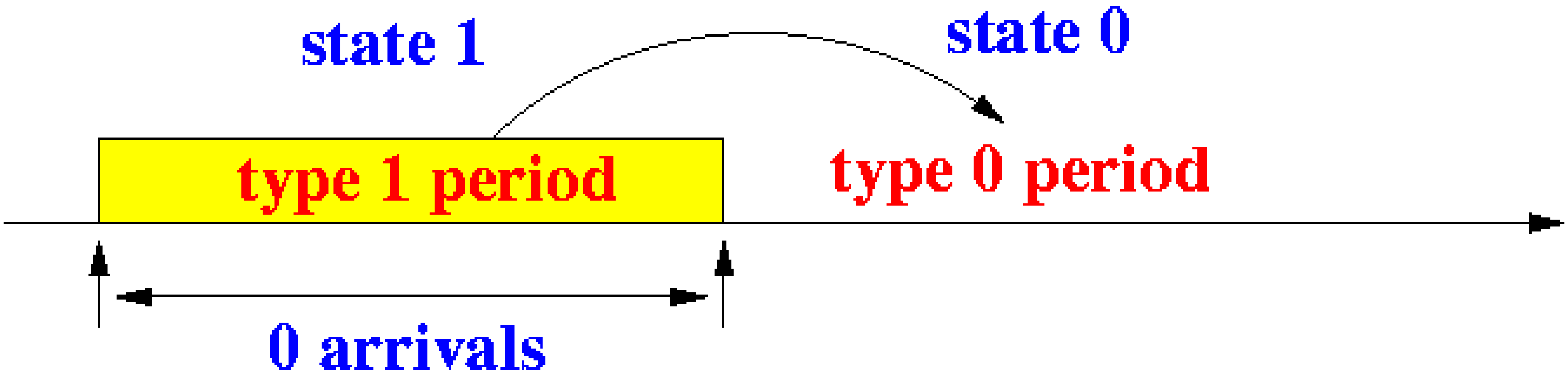
$$\pi_1 = \frac{p_{10} + p_{11}}{1 + p_{10}}$$

$$\pi_2 = \frac{1 - p_{10} - p_{11}}{1 + p_{10}}$$

Now all we still need to do is compute p_{10} and p_{11}

Computing the transition probability p_{10}

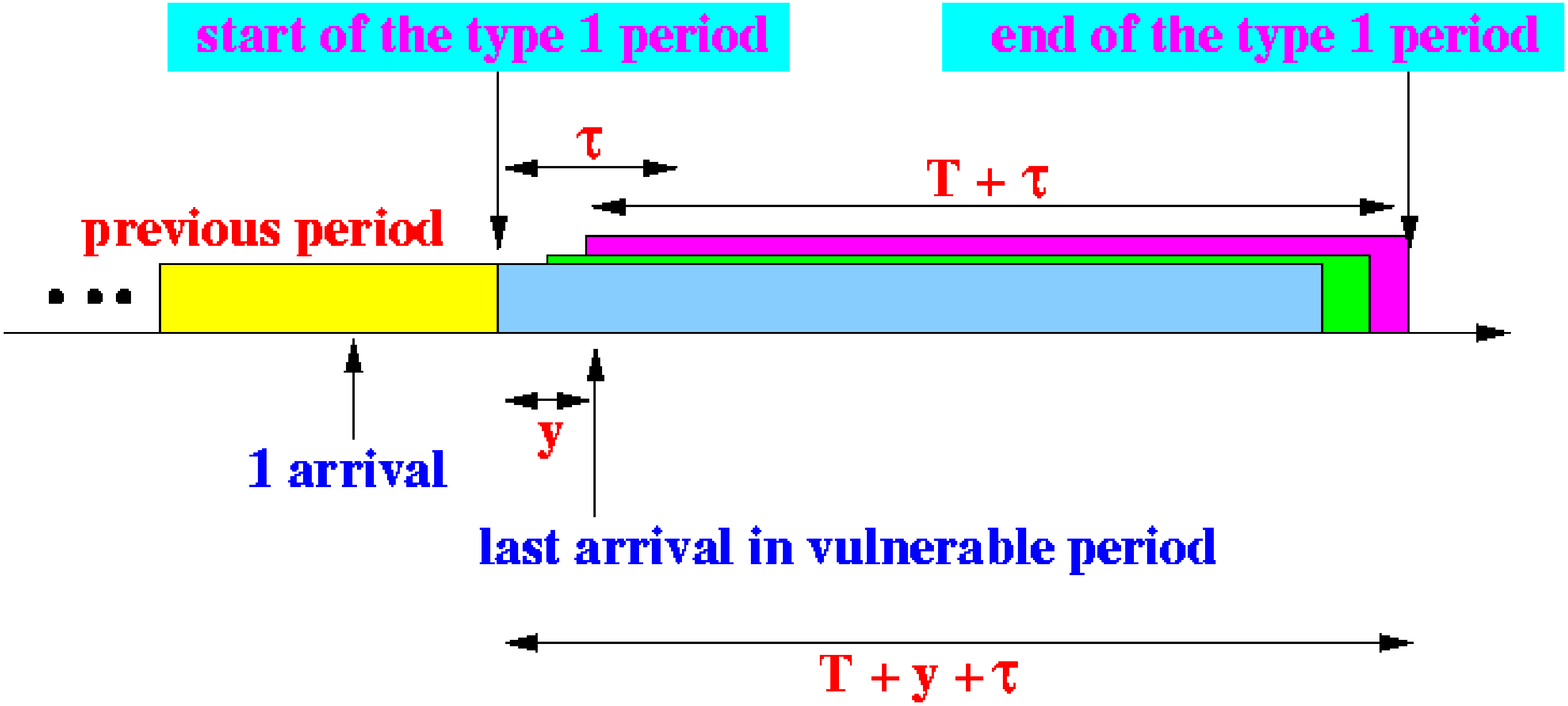
- **Transition from state 1 to state 0:**



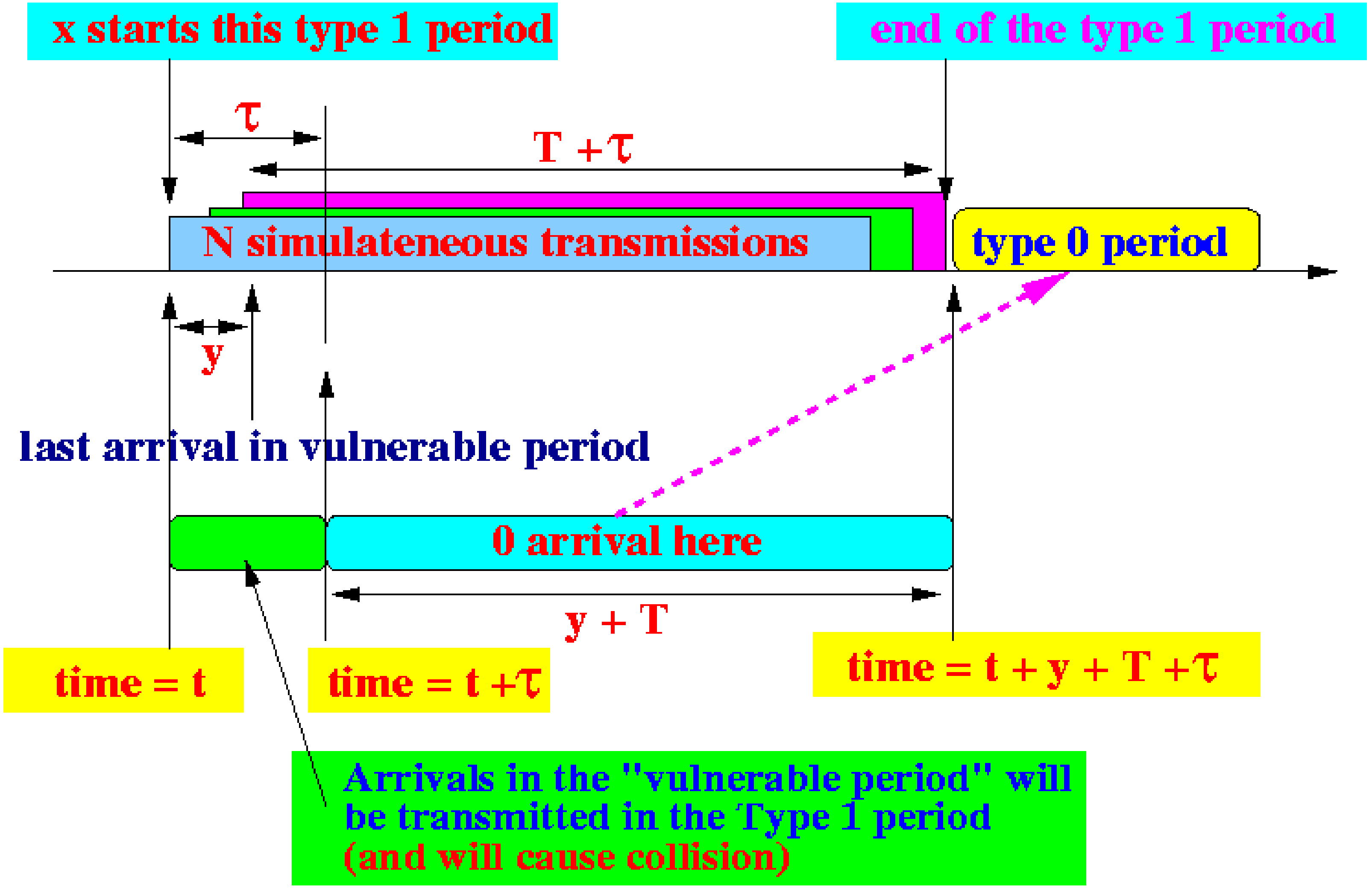
- The system moves **from state 1 to state 0 (idle)** when there are **0 arrivals** within the **type 1 period**

- The **number of arrivals** in the **type 1 period** depends on the ***duration* (length) of the type 1 period**

- To obtain the **duration** of the **type 1 period**, we must look at the **anatomy** of a **type 1 transmission period**:



Suppose a **type 1 period** is started at **time t** by the packet transmission **x**:



Observation 1:

- **Arrivals** that arrive ***before*** **time t + τ**

  will ***not*** **detect** the **starting transmission x**
- **Consequently**, they will **schedule themselves** for transmission ***within*** **the type 1 period** (and they will cause a **collision**)

- **Arrivals** that arrive ***after*** **time t + τ**

  will **detect** the **starting transmission x**
- **Consequently**, they will **schedule themselves** for transmission ***after*** **the type 1 period**

Observation 2:

- The **total length** of the **type 1 period** is $\mathbf{T + \tau + y}$
- (**y** is the **time of the *last* arriving packet**

  before **deadline** $\boldsymbol{\tau}$)
- In order to **transit** to a **type 0 (idle) period**, there **must be zero (0) arrivals** between:

$$[\mathbf{t + \tau, t + y + T + \tau})$$

i.e., there **must be zero (0) arrivals** in **(T + y) sec**

- **Thus:** the **probability** that a **state 1 period** will transits into a **type 0 period** is equal to:

- p_{10} = P[0 arrivals in (T + y) sec]

- If **y were** a **constant**, the solution would be **simple**:

-

$$\frac{(g(T + y))^0}{0!} e^{-g(T + y)} \qquad \text{(Poisson arrivals)}$$

BUT, y is a random variable

- The **technique** used to compute **P[0 arrivals in (T + y) sec]** is the **Law of Total Probability**

- (We divide the **range "[0, τ]"** into **individual time instances u**)

$p_{10} = \sum_{u=0}^{\tau} P[\text{ 0 arrivals in } T + y \mid y = u\,] \times P[\, y = u\,]$ (Law of Total Probability)

$= \sum_{u=0}^{\tau} P[\text{ 0 arrivals in } T + u\,] \times P[\, y = u\,]$

- The sum is in fact an integral because the random variable y is a continuous random variable:

- $p_{10} = \int^{\tau}_{u=0} P[\ 0 \text{ arrivals in } T + u\ ] \times f_y(u)\ du$

The pdf $f_y(t)$ has been discovered before !
From the discussion on the non-persistent CSMA protocol, we got:

$f_y(t)$ = the probability density function of y
$= e^{-g\tau} \times \delta(t) + g \times e^{-g(\tau-t)}$

Therefore:

$$p_{10} = \int^{\tau}_{u=0} P[\ 0 \text{ arrivals in } T + u\ ] \times (e^{-g\tau} \times \delta(u) + g \times e^{-g(\tau-u)})\ du$$

The probability of k arrivals in t sec in a Poisson process is:

$$P[\ k \text{ arrivals in } t \text{ sec}\ ] = \frac{(gt)^k}{k!} e^{-gt}$$

So the probability of 0 arrivals in T+u sec in a Poisson process is:

$$P[\ 0 \text{ arrivals in } T{+}u \text{ sec}\ ] = \frac{(g(T+u))^0}{0!} e^{-g(T+u)} = e^{-g(T+u)}$$

Therefore:

(Split off the discontinuous point u=0)

$$p_{10} = \int^{\tau}_{u=0} e^{-g(T+u)} \times (e^{-g\tau} \times \delta(u) + g \times e^{-g(\tau-u)})\, du$$

$$= e^{-g(T+0)} \times e^{-g\tau} + \int^{\tau}_{u=0+} e^{-g(T+u)} \times g \times e^{-g(\tau-u)}\, du$$

$$= e^{-g(T+\tau)} + \int^{\tau}_{u=0+} e^{-g(T+u)} \times g \times e^{-g(\tau-u)}\, du$$

- Computing the integral we get -

Therefore:

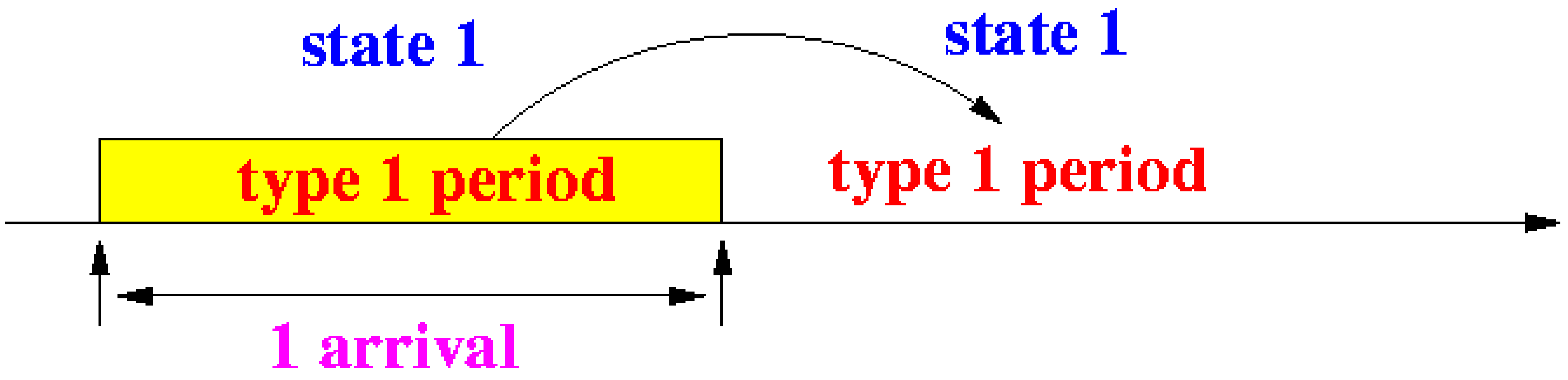
$$p_{10} = e^{-g(T+\tau)} + g\tau\, e^{-g(T+\tau)}$$

$$= (1 + g\tau)\, e^{-g(T+\tau)}$$

- We now only have to compute $\mathbf{p_{11}}$ before we can find the **throughput formula** for the **1-persistent CSMA protocol**....

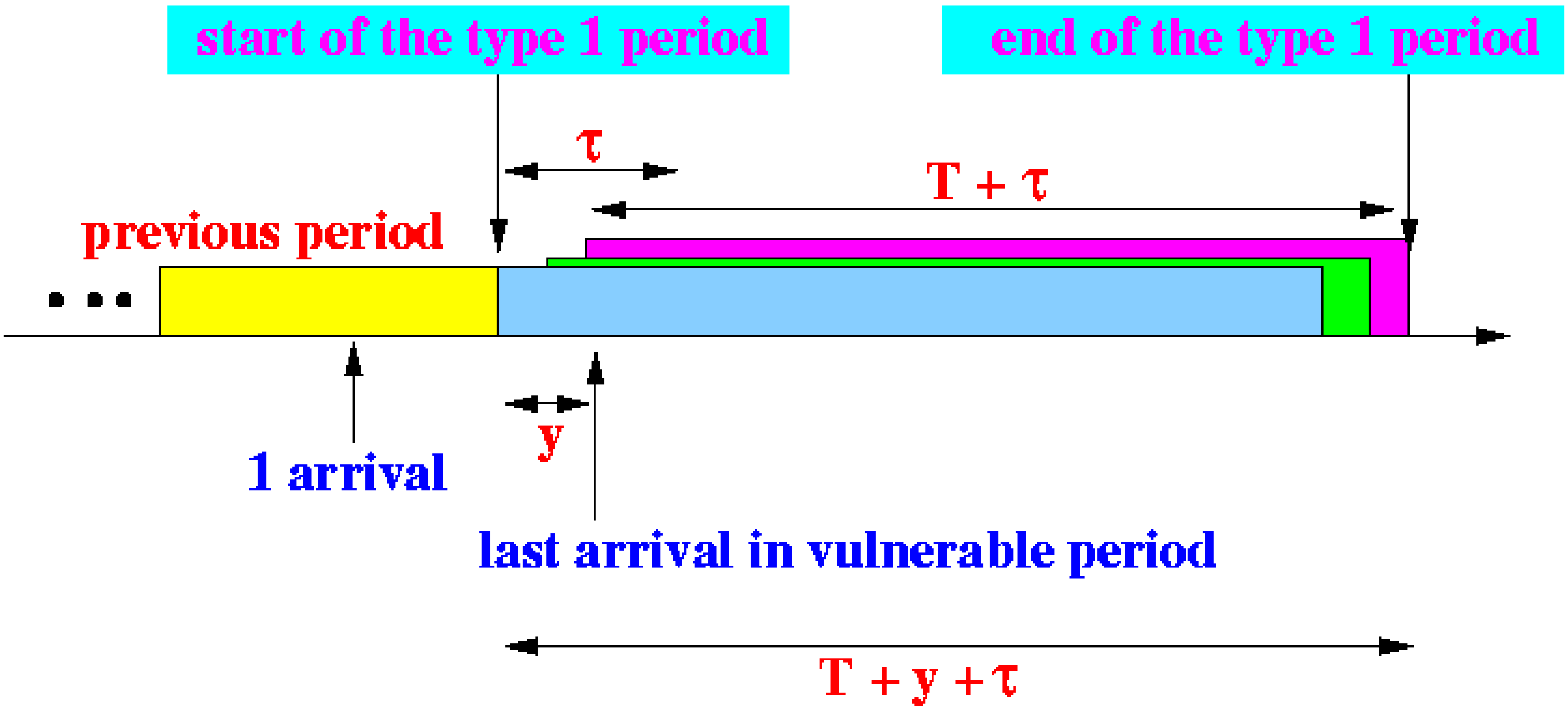
Computing the transition probability p_{11}

- **Transition from state 1 to state 1:**
-

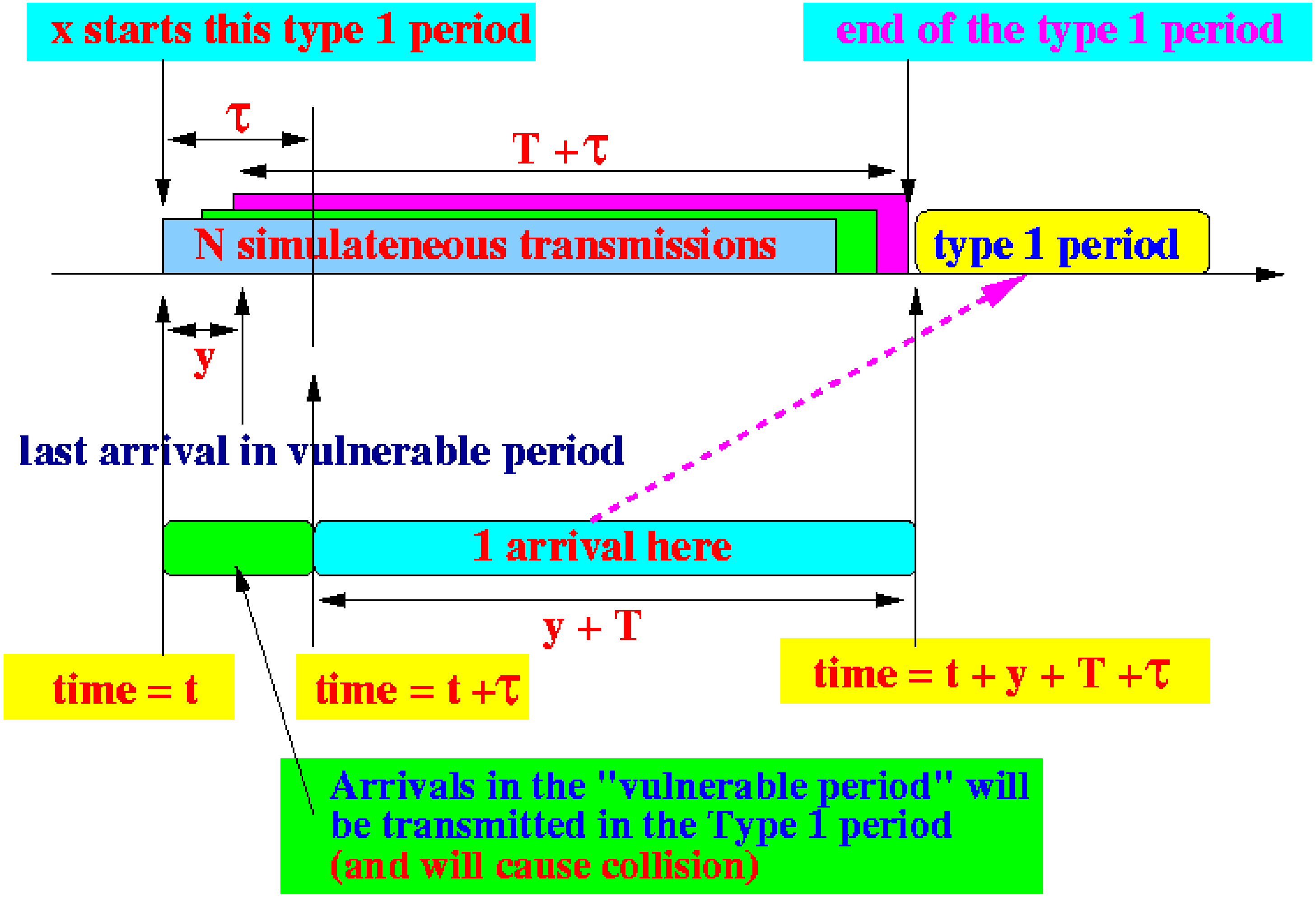


- The system moves **from state 1 to state 1** when there is ***exactly*** **1 arrival** in the **type 1 period**
- **Again**, the **number of arrivals** in the **type 1 period** depends on the ***duration*** **(length) of the type 1 period**

To obtain the **duration** of the **type 1 period**, we must look at the **anatomy** of a **type 1 transmission period**:



Suppose a **type 1 period** is started at **time t** by the packet transmission **x**:



Observation 1:

- **Arrivals** that arrive ***before*** **time t + τ**

  will ***not*** **detect** the **starting transmission x**
- **Consequently**, they will **schedule themselves** for transmission ***within*** **the type 1 period** (and they will cause a **collision**)
- **Arrivals** that arrive ***after*** **time t + τ**

  will **detect** the **starting transmission x**
- **Consequently**, they will **schedule themselves** for transmission ***after*** **the type 1 period**

Observation 2:

- The **total length** of the **type 1 period** is T + **τ** + y
- (**y** is the **time of the** ***last*** **arriving packet** before **deadline τ**)
- In order to **transit** to a **type 1 period**, there **must be one (1) arrival** between:

  [t + τ, t + y + T + τ)

  i.e., there **must be one (1) arrivals** in **(T + y) sec**

Thus: the **probability** that a **state 1 period** will transits into a **type 0 period** is equal to:

- p_{11} = P[1 arrival in (T + y) sec]

- **Again**, if **y were** a **constant**, the solution would be **simple**:

-

$$\frac{(g(T + y))^1}{1!} e^{-g(T + y)} \qquad \text{(Poisson arrivals)}$$

BUT, y is a random variable

- We will need use the **Law of Total Probability** to compute **P[0 arrivals in (T + y) sec]**:

- (We divide the **range "[0, τ]"** into **individual time instances u**)

-

$$p_{11} = \sum^{\tau}_{u=0} P[\ 1 \text{ arrival in } T + y \mid y = u\ ] \times P[\ y = u\ ]$$

(Law of Tatal Probability)

$$= \sum^{\tau}_{u=0} P[\ 1 \text{ arrival in } T + u\ ] \times P[\ y = u\ ]$$

The sum is in fact an integral because the random variable y is a continuous random variable:

$$p_{11} = \int^{\tau}_{u=0} P[\ 1 \text{ arrival in } T + u\ ] \times f_y(u)\, du$$

Again, the difference is the variable u is NOT a random variable

From the discussion on the non-persistent CSMA protocol, we got:

$f_y(t)$ = the probab. density function of y

$$= e^{-g\tau} \times \delta(t) + g \times e^{-g(\tau - t)}$$

Therefore:

$$p_{11} = \int^{\tau}_{u=0} P[\text{ 1 arrival in } T + u\ ]$$

$$\times\ (e^{-g\tau} \times \delta(u) + g \times e^{-g(\tau-u)})\ du$$

The probability of k arrivals in t sec in a Poisson process is:

$$P[\ k \text{ arrivals in } t \text{ sec}\ ] = \frac{(gt)^k}{k!} e^{-gt}$$

So the probability of 1 arrival in T+u sec in a Poisson process is:

$$P[\ 1 \text{ arrivals in } T+u \text{ sec}\ ] = \frac{(g(T+u))^1}{1!} e^{-g(T+u)} = g(T+u)e^{-g(T+u)}$$

Therefore:

$$p_{11} = \int_{u=0}^{\tau} g(T+u)\, e^{-g(T+u)} \times (e^{-g\tau} \times \delta(u) + g \times e^{-g(\tau-u)})\, du$$

(Split off the discontinuous point u=0)

$$= g(T+0)\, e^{-g(T+0)} \times e^{-g\tau} +$$

$$\int_{u=0+}^{\tau} g(T+u)\, e^{-g(T+u)} \times g \times e^{-g(\tau-u)}\, du$$

$$= gT\, e^{-g(T+\tau)} \quad +$$

$$\int^{\tau}_{u=0+} g(T+u)\, e^{-g(T+u)} \times g \times e^{-g(\tau-u)}\, du$$

Solving it …

Therefore: (Take $g\ e^{-g(T+\tau)}$ out)

$$p_{11} = gT\ e^{-g(T+\tau)} + 1/2\ \tau\ (2\ T + \tau)\ g^2\ e^{-g(T+\tau)}$$

$$= gT\ e^{-g(T+\tau)} + \tau\ (T + \tau/2)\ g^2\ e^{-g(T+\tau)}$$

$$= g\ e^{-g(T+\tau)}\ (T\ +\ \tau\ (T + \tau/2)\ g\ )$$

$$= g\ e^{-g(T+\tau)}\ (T + g\tau(T + \tau/2))$$

Finally: throughput expression for 1-persistent CSMA

$$S = \frac{gT\, e^{-g(T+2\tau)} \left[1 + gT + g\tau(1 + gT + g\tau/2) \right]}{g(T + 2\tau) - (1 - e^{-g\tau}) + (1 + g\tau)e^{-g(T+\tau)}}$$

The *normalized* throughput expression for 1-persistent CSMA

- Often, we see the **normalized form** (normalized by **packet transmission time T**)
- **Substitutions used to normalize the throughput expression:**

g = offered packet rate per sec
G = gT = offered packet rate per packet transmission time T
τ = end to end propagation delay in sec
a = τ / T = end to end propagation delay in packet transmission time T

A useful derived equality:

$$a = \tau/T \Leftrightarrow aT = \tau \Leftrightarrow agT = g\tau \Leftrightarrow aG = g\tau$$

Normalized throughput expression:

$$S = \frac{gT\, e^{-g(T+2\tau)}\ [\ 1 + gT + g\tau(1 + gT + g\tau/2)\ ]}{g(T + 2\tau) - (1 - e^{-g\tau}) + (1 + g\tau)e^{-g(T+\tau)}}$$

(Subst: G = gT)

$$= \frac{gT\, e^{-(gT+2g\tau)}\ [\ 1 + gT + g\tau(1 + gT + g\tau/2)\ ]}{(gT + 2g\tau) - (1 - e^{-g\tau}) + (1 + g\tau)e^{-(gT+g\tau)}}$$

$$= \frac{G\, e^{-(G+2g\tau)}\ [\ 1 + G + g\tau(1 + G + g\tau/2)\ ]}{(G + 2g\tau) - (1 - e^{-g\tau}) + (1 + g\tau)e^{-(G+g\tau)}}$$

(Subst: aG = gτ)

$$= \frac{G\, e^{-(G+2aG)}\ [\ 1 + G + aG(1 + G + aG/2)\ ]}{(G + 2aG) - (1 - e^{-aG}) + (1 + aG)e^{-(G+aG)}}$$

$$= \frac{G\, e^{-G(1+2a)}\ [\ 1 + G + aG(1 + G + aG/2)\ ]}{G(1 + 2a) - (1 - e^{-aG}) + (1 + aG)e^{-G(1+a)}}$$

(Rewrite nicer...)

This is the **classic form** for the **throughput** of **1-persistent CSMA**

$$= \frac{G\, e^{-G(1+2a)}\,[\,1 + G + aG(1 + G + aG/2)\,]}{G(1 + 2a) - (1 - e^{-aG}) + (1 + aG)e^{-G(1+a)}}$$

Summary

1 **Markov Process:** Transition Probability Diagram

- All. : P00, P01, P02, P10, P11, P12, P20, P21, P22
- But : P01, P10, P11, P12, P20, P21, P22

2

- Stationary Probabilities: $\pi 0$, $\pi 1$, $\pi 2$
- Length: $\underline{T0}$, $\underline{T1}$, $\underline{T2}$
- So $S = (\pi 0 \times s(\underline{T0}) + \pi 1 \times s(\underline{T1}) + \pi 2 \times s(\underline{T2}))$ / Total

- Find $\pi 0$, $\pi 1$, $\pi 2$, $\underline{T0}$, $\underline{T1}$, $\underline{T2}$

Summary

Finding: $\pi 0$, $\pi 1$, $\pi 2$

Note that: T1 = T2

So, P10 = P20, P11= P21

Using these we can simplify the stationary equations and get $\pi 0$, $\pi 1$, $\pi 2$ as function of only P10 and P11

Finally we find these two through total probability law.

Performance graphs

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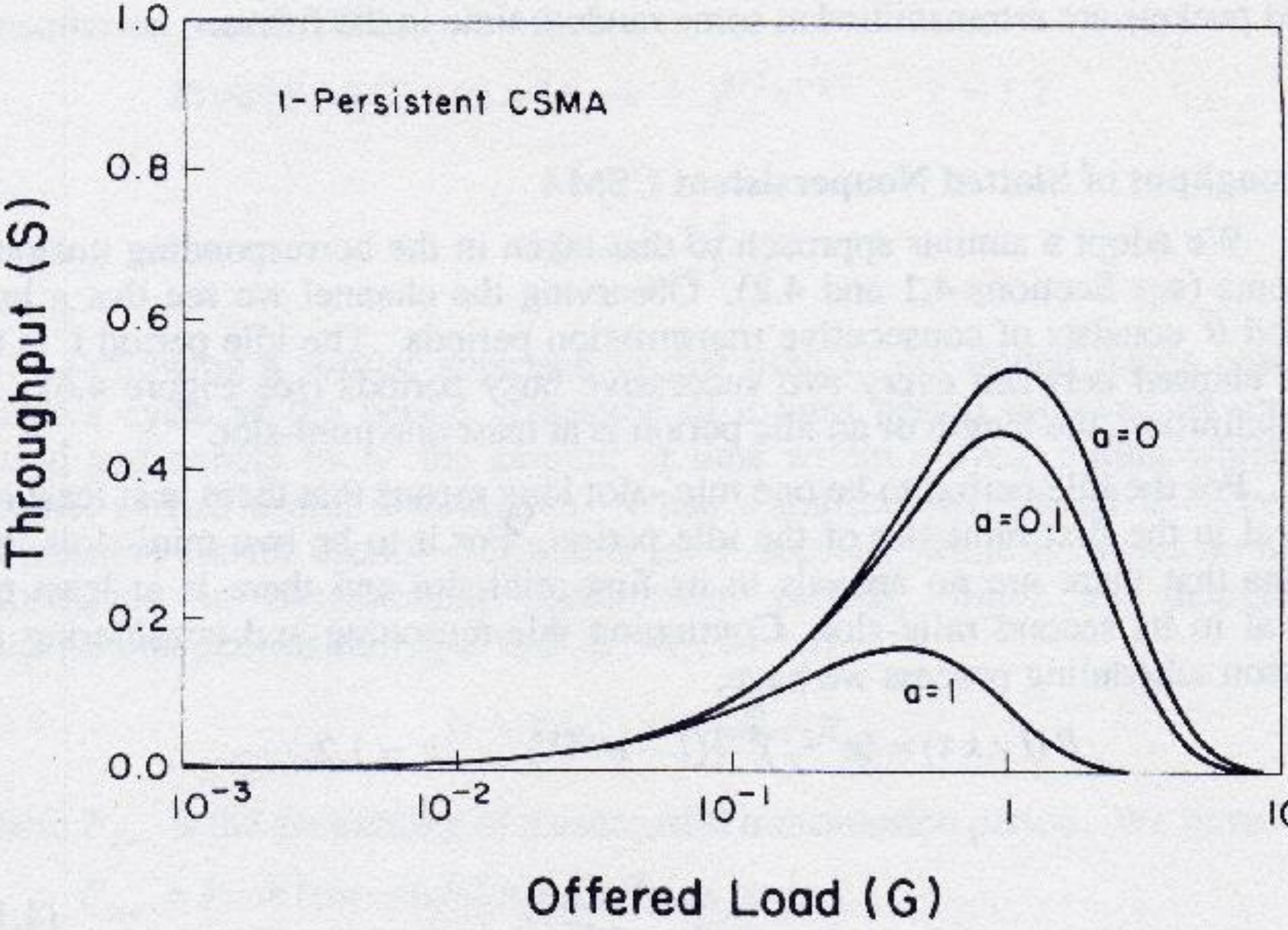

Figure 4.5: Throughput-Load of 1-Persistent CSMA

Comparing with Non-persistent CSMA...

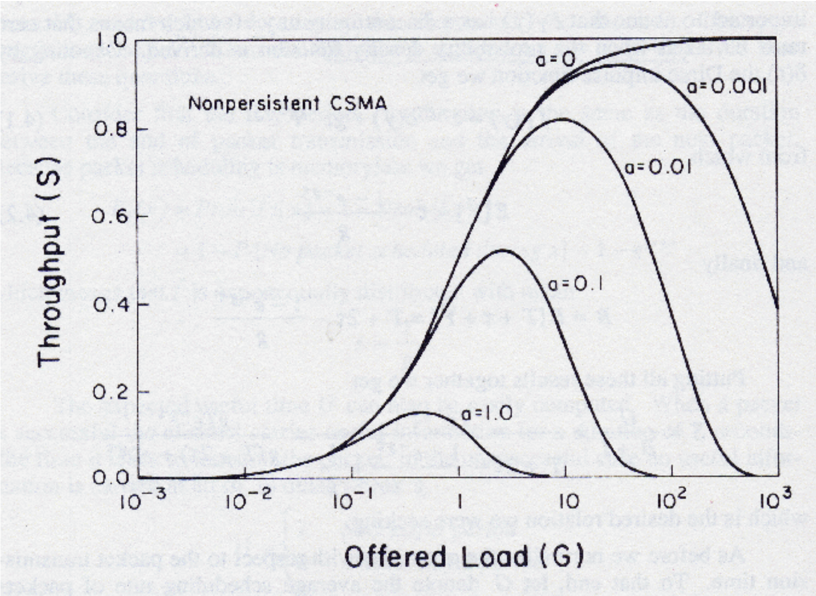

Figure 4.2: Throughput-Load of Nonpersistent CSMA

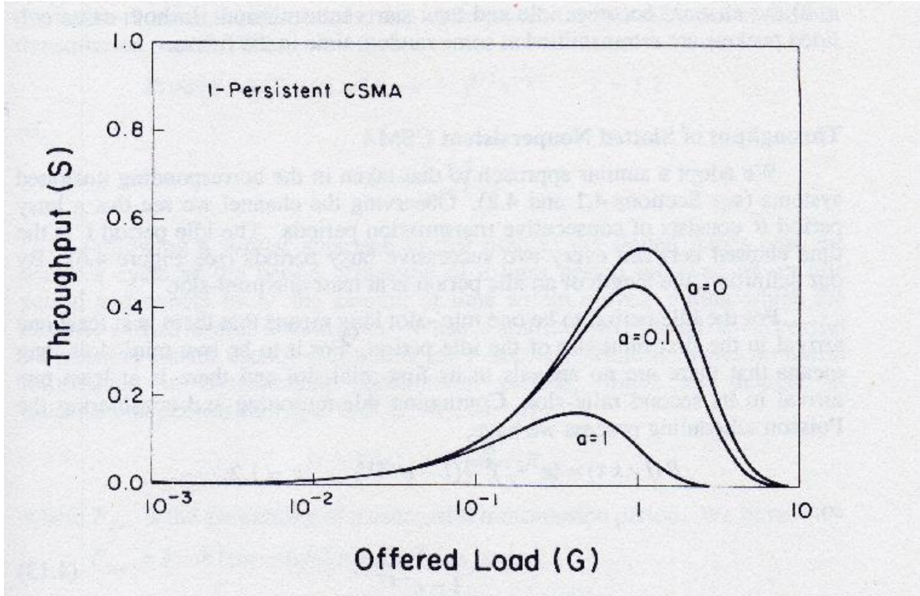

Figure 4.5: Throughput-Load of 1-Persistent CSMA

- **1-persistent CSMA** performs **slightly better** at **lighter loads**